

ECE 528/L

Robotics and Embedded Systems with Lab

Final Project Project Report

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Introduction:

The goal of this final project is to energize the TF-Luna LiDAR sensor and have it gather data through its IR sensor. The data bytes will be collected and sent over to the ESP32 MCU. And the serial data will then be sent over to the laptop/PC peripheral which will then be used to display visualizations of the collected sensor data.

Components:

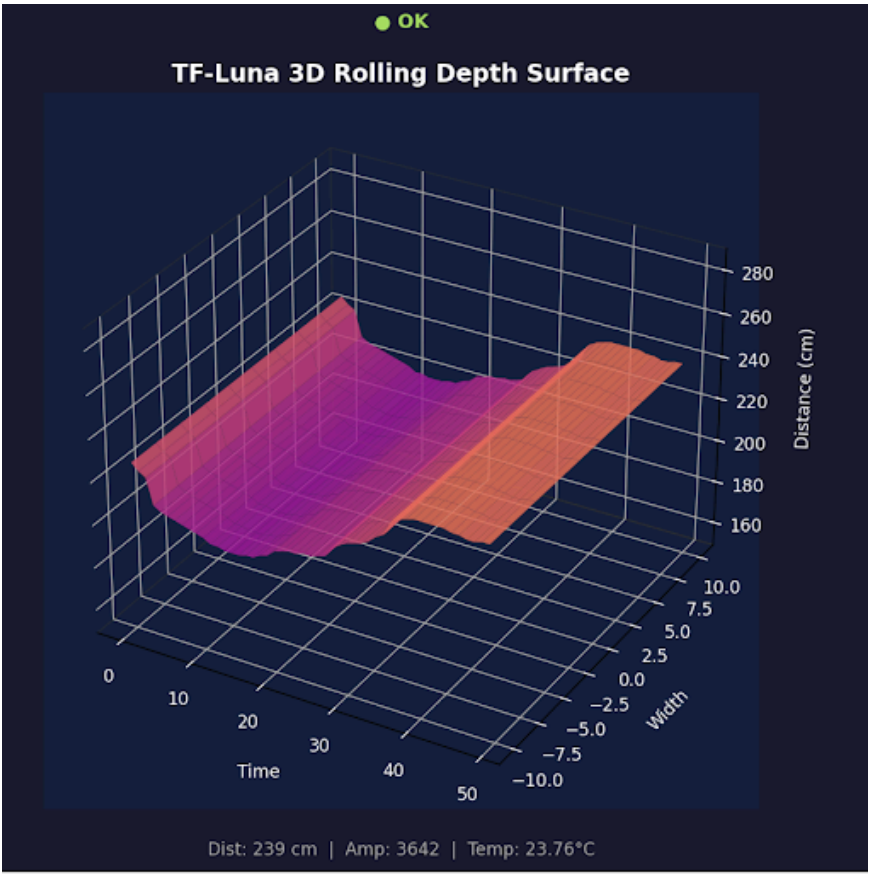
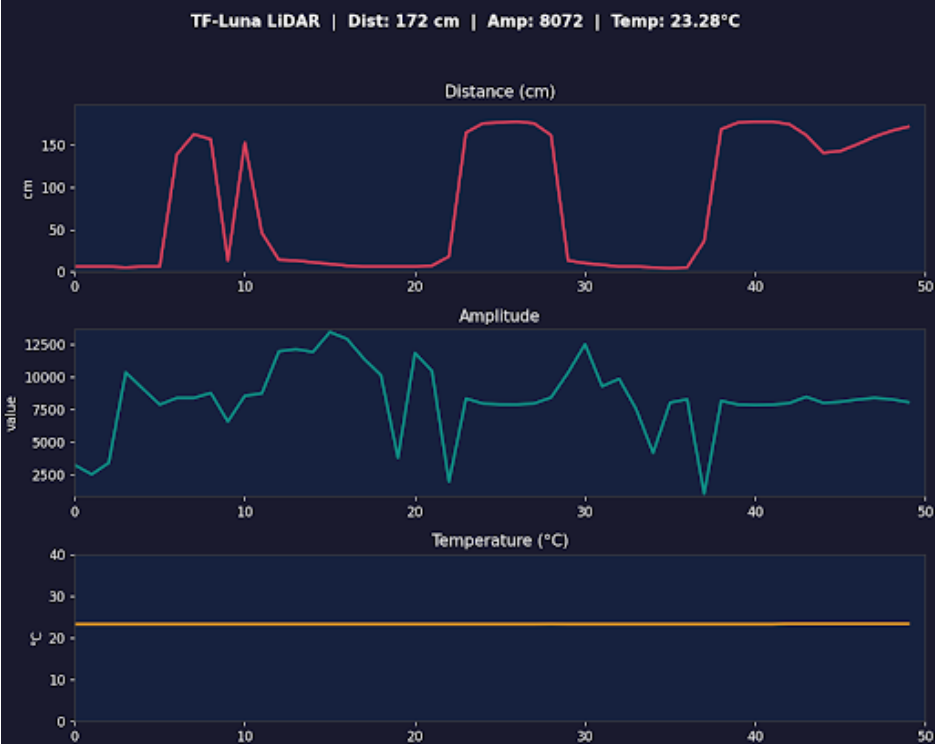
- Laptop/PC
- USB-C cable
- ESP32-S3-DevKitC1
- TF-Luna LiDAR sensor

Analysis and Results:

The following image below verifies that the lidar sensor is energized and is connected properly. This is important to make sure that the data is being transmitted over to the ESP32.



Below are a few examples of graph visualizations plotting and showing the gathered sensor data. It is the result of the program processing several data points and presenting them in a fashion that is understandable for the user to analyze.



Author Contributions:

Joaquin Tapia:

- Performed project
- Provided project report

Conclusion:

The project was successful in achieving its goal, creating visualizations of the gathered lidar sensor data. It shows that the lidar sensor was properly connected and the data was successfully transmitted, from the sensor to the MCU, to then the laptop/PC peripheral. The combination of C++ and python then resulted in a successful output of graphs that portray information about what and where the sensor is collecting data. Unfortunately, due to lack of time, the lidar sensor could be used for complex applications, such as depicting a live 3D surrounding of its environment through the use of servo motors.

Github link:

<https://github.com/joaquin-tapia-ee/ECE-528-Final-Project>